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300118(24)

B. E. (PQ) (First & Second Semester) Examination 2020

(New Scheme)

(AEI, Bio Tech, Chem., Civil, CSE, Elect., EEE, EI, Et & T, IT, Mech., Mining, Metallurgy, Mechatronics, Production, Automobile Branch)

ELEMENTS of ELECTRICAL ENGINEERING

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Part (a) of each question is compulsory. Attempt any two parts from (b), (c) & (d) of each question.*

Unit - I

1. (a) What is active and passive elements? 2

(b) Calculate the direction and magnitude of the current through 5 Ω resistor between points A and B of figure 1(b) using nodal voltage method. 7

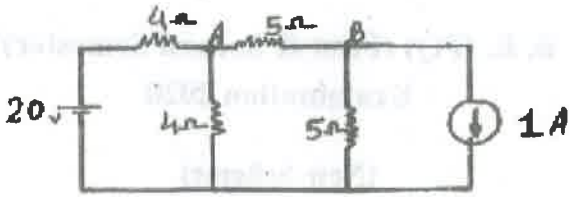

Fig.-1 (b)

(c) Use superposition theorem to find current I in the circuit shown in figure 1 (c). 7

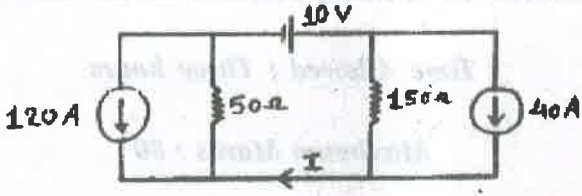

Fig.-1 (c)

(d) Derive the condition for maximum power transfer to load R_L from a voltage source having internal resistance R_s. 7

Unit - II

2. (a) Define form factor and peak factor. 2

(b) Three impedances Z_1, Z_2, Z_3 are connected in series across 100 volt, 50Hz supply if

$$Z_1 = (4 + j3)\Omega,\ Z_2 = (6 - j8)\Omega,\ Z_3 = 4\Omega$$

Find :

(i) Current

(ii) Power factor of circuit

(iii) Power absorbed by the circuit 7

(c) Two circuits the impedance of which are given by $Z_1 = (15 + j12)\Omega$ and $Z_2 = (8 + j5)\Omega$ are connected in parallel if the potential difference across one of the impedance is (250 + j0)V, calculate :

(i) Total current

(ii) Total Power

(iii) Overall Power Factors 7

(d) In a series parallel circuit, the parallel branches A & B are in series with C. The impedances are

$$Z_A = (4 + j3)\Omega,\ Z_B = (10 - j7)\Omega,$$

$$Z_C = (6 + j5)\Omega$$

If the voltage applied to the circuit is 200V, 50Hz. Calculate :

(i) Current I_A, I_B, I_C, i.e. branch currents.

(ii) Total power factor for the whole circuit. 7

Unit - III

3. (a) What is leakage flux? 2

(b) Establish relationship between line and phase voltage in a balanced 3-phase star connection. 7

(c) A delta connected balanced 3-phase load is supplied from a 3-phase, 400V supply. The line current is 20A and the power taken by load is 10,000W. Find :

(i) Impedance is each branch

(ii) If same load is connected in star across same supply find line current and power consumed. 7

(d) A mild steel ring of 30 cm mean circumference has a cross sectional area of 6 cm^2 and has a winding of 500 turns on it. The ring is cut through at a point so as to provide an airgap of 1 mm in the magnetic circuit. It is found that a current of 4A in the winding, produces of flux density of 1T

in the airgap. Find relative permeability of mild steel. 7

Unit - IV

4. (a) What is the necessity of starter in DC motor? 2

(b) (i) Derive an expression for the emf induced in a transformer winding. 4

(ii) Derive emf equation in DC machine. 3

(c) The efficiency of a 400 kVA, single phase transformer is 98·77% when delivering full load at 0.8 power factor and 99·13% at half load and unity power factor. Calculate Iron loss and full load copper loss. 7

(d) The armature of a 12 pole dc shunt generator has 50 slots and is wave wound with 12 conductors per slot. The generator is running at a speed of 625 rpm and supplies a resistive load of 15 Ω at a terminal voltage of 300V. The armature resistance is 0·5 Ω and field resistance is 60 ohm. Find armature current, generated emf and flux per pole. 7

Unit - V

5. (a) How instruments are classified? 2

(b) Explain in brief essential requirements of electrical instruments. 7

(c) Explain with neat sketch, the construction & working of PMMC instruments. 7

(d) A PMMC instrument gives full scale deflection at a potential difference of 75 mV and a current of 25 mA, How can we use the instrument to measure. 7

(i) Current upto 100 A

(ii) Voltage upto 750 V.